

BRANCH COUNTY AP, MI, USA

WMO: 725414

Lat: 41.933N

Lon: 85.053W

Elev: 959

StdP: 14.19

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 04886

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.2	4.9	-7.7	3.8	1.0	-1.8	5.2	6.8	24.0	25.1	21.1	25.1	5.7	230	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	89.6	72.8	86.5	71.5	83.9	70.5	77.2	83.6	75.3	82.0	73.8	80.3	8.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.1	136.3	80.7	73.1	127.3	78.6	72.1	122.9	77.5	41.4	82.8	39.4	81.7	37.9	80.2	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.0	18.7	16.7	-8.4	92.7	5.5	3.1	-12.4	94.9	-15.6	96.7	-18.7	98.5	-22.7	100.8		
			-8.9	79.7	5.3	2.3	-12.7	81.3	-15.8	82.7	-18.8	83.9	-22.6	85.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	24.5	26.5	36.8	48.9	59.9	68.8	72.0	70.0	64.1	52.1	40.7	29.8	
	DBStd	19.11	11.52	11.18	11.14	9.36	8.72	6.63	5.49	5.25	7.63	9.06	9.59	10.37	
	HDD50	3051	790	660	433	130	16	0	0	0	2	84	306	630	
	HDD65	6299	1254	1078	875	488	208	35	7	15	108	411	728	1092	
	CDD50	2919	1	3	24	97	322	565	682	620	424	150	28	3	
	CDD65	691	0	0	1	6	49	150	224	170	80	11	0	0	
	CDH74	5839	0	0	11	53	434	1282	1982	1296	694	87	0	0	
	CDH80	1678	0	0	1	7	96	400	635	321	205	13	0	0	
Wind	WSAvg	7.0	8.6	8.4	8.2	8.8	7.2	5.9	4.9	4.5	5.2	6.9	7.8	8.3	
Precipitation	PrecAvg	34.4	1.8	1.5	2.4	3.4	3.9	3.4	3.9	3.6	3.2	2.6	2.6	2.3	
	PrecMax	44.3	5.2	5.3	4.6	6.7	10.9	6.9	9.6	9.3	6.2	8.4	5.6	4.6	
	PrecMin	20.6	0.2	0.3	0.4	0.3	1.0	0.7	0.7	0.4	0.0	0.4	0.3	0.5	
	PrecStd	5.3	1.2	1.0	1.1	1.5	1.8	1.7	2.1	2.1	1.5	1.5	1.3	1.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.9	60.6	74.6	80.7	86.8	91.2	93.2	90.1	89.9	81.8	70.0	60.6	
		MCWB	53.7	54.2	62.2	63.7	68.0	72.1	74.1	74.9	70.4	68.3	58.7	57.2	
	2%	DB	50.1	54.1	65.6	73.8	82.3	88.2	89.6	86.4	86.0	75.7	63.7	54.1	
		MCWB	46.9	50.5	57.5	60.1	66.7	71.3	74.3	72.8	70.1	63.4	55.4	51.5	
	5%	DB	44.8	46.8	59.3	70.4	79.3	84.3	86.3	83.6	81.6	71.8	60.2	47.9	
		MCWB	41.9	43.5	52.6	57.2	66.0	69.1	72.5	71.8	68.2	62.3	53.1	45.2	
	10%	DB	39.2	41.1	54.4	65.6	74.9	81.4	83.6	81.1	77.6	67.2	55.2	43.2	
		MCWB	36.9	37.7	48.2	55.0	63.7	69.1	71.6	70.7	66.4	59.5	49.7	40.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.4	56.9	63.1	68.7	73.6	76.7	81.0	79.9	76.1	71.0	61.3	57.4
			MCDB	56.8	58.3	73.9	76.7	80.7	84.5	86.6	86.1	83.4	77.5	66.2	60.0
2%		WB	47.8	50.3	58.3	63.8	70.5	74.8	77.7	76.8	73.4	67.5	58.1	51.4	
		MCDB	48.7	52.9	64.2	69.7	77.6	82.6	84.0	82.5	80.3	72.5	62.5	53.0	
5%		WB	41.9	43.7	54.3	59.8	68.2	73.1	75.7	74.6	71.5	64.2	54.4	45.6	
		MCDB	44.4	47.3	58.5	66.9	75.8	80.5	82.5	80.2	77.8	68.2	58.6	46.9	
10%		WB	37.4	37.8	48.3	56.2	66.1	71.2	74.1	72.7	69.2	60.8	50.6	41.0	
		MCDB	39.2	40.7	53.1	64.3	72.6	78.4	80.7	78.2	74.2	65.9	54.9	43.4	
Mean Daily Temperature Range		5% DB	MDBR	13.3	14.6	16.8	20.1	19.6	19.0	19.6	19.3	20.6	18.3	15.0	12.0
			MCDBR	16.6	19.2	24.2	25.2	23.1	22.3	21.7	21.8	24.2	23.4	20.3	17.1
	MCWBR		14.6	15.8	16.4	14.4	12.5	9.3	9.6	10.6	10.8	12.9	14.4	14.8	
	5% WB	MCDBR	15.6	18.3	21.8	21.3	19.7	17.8	17.7	17.4	19.1	18.5	18.2	16.0	
		MCWBR	14.5	16.1	16.4	15.4	12.9	9.7	10.1	10.3	10.3	13.3	14.8	15.0	
Clear-Sky Solar Irradiance	taub	0.300	0.350	0.351	0.392	0.425	0.441	0.430	0.437	0.391	0.350	0.318	0.302		
	taud	2.373	2.193	2.333	2.243	2.195	2.191	2.230	2.203	2.315	2.425	2.462	2.405		
	Ebn at Noon	268	266	283	278	271	266	267	261	266	266	260	256		
	Edh at Noon	26	36	36	42	45	46	43	43	36	29	24	23		
All-Sky Solar Radiation	RadAvg	464	740	1116	1463	1695	1911	1938	1679	1366	885	554	384		
	RadStd	46	103	108	133	161	124	95	84	119	87	78	43		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page